

National-Scale Healthcare Management System (HMS)

1. Executive Summary

The National-Scale Healthcare Management System (HMS) will unify patient care, provider operations, insurance processing, billing, inventory, and public health reporting across public and private healthcare facilities nationwide. This BRD defines the **business goals, scope, processes, and detailed business requirements** to guide solution design, procurement, implementation, and user acceptance. It deliberately avoids technical implementation details.

Business Outcomes (12–24 months):

- Single longitudinal health record for $\geq 80\%$ of the population.
 - 30% reduction in average outpatient waiting time and 20% improvement in bed turnover.
 - $\geq 95\%$ on-time public health reporting for notifiable conditions.
 - $\geq 15\%$ reduction in claim rejection rates and $\geq 20\%$ faster revenue cycle closure.
 - Full auditability of clinical, financial, and administrative actions.
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2. Business Context

2.1 Current Challenges

- Fragmented patient identities and duplicate records.
- Manual handoffs across registration → clinical → pharmacy → billing → insurance.
- Delayed or incomplete lab/radiology results and critical value escalations.
- Inconsistent claim adjudication documentation and payer rules.

- Poor stock visibility (drug expiries, shortages), leading to care delays.
- Inconsistent reporting to national disease registries.

2.2 Target Future State

- One patient view across facilities with consent-driven access.
 - Standardized, guided workflows from appointment to discharge and follow-up.
 - Time-bound notifications for critical events (e.g., critical lab results).
 - Transparent, rules-driven billing and claims; payer collaboration.
 - Inventory visibility from national to facility level with proactive replenishment.
 - Timely, privacy-preserving public health surveillance and analytics.
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3. Goals & Objectives (SMART)

- **G1 Access & Continuity:** Establish a lifelong patient profile with consent preferences; **≥ 98%** successful patient retrieval at point of care.
- **G2 Service Efficiency:** Reduce outpatient visit cycle time by **30%** within 12 months of go-live at a facility.
- **G3 Safety & Quality:** Ensure **100%** acknowledgment of critical lab/imaging results within **30 minutes** of verification.
- **G4 Financial Integrity:** Achieve **≤ 5%** claim rejection rate nationally by month 18.
- **G5 Public Health:** Report notifiable diseases **within 60 minutes** of confirmation in **≥ 95%** of cases.
- **G6 Governance:** Maintain complete, tamper-evident audit histories for **100%** sensitive actions.

4. Scope

4.1 In Scope

Patient administration, appointments & encounters, clinical documentation, medications (e-prescription & dispensing), diagnostics (lab & imaging), insurance eligibility & claims, billing & payments, inventory & procurement, public health registries, analytics & dashboards, patient portal, provider registry, and national data exchanges (identity, drugs, diseases).

4.2 Out of Scope (Phase 1)

Advanced genomics, tele-surgery, third-party wellness marketplaces, cross-border medical tourism billing, and research data warehouses (non-operational analytics).

5. Stakeholders & Personas

- **Patients & Caregivers:** Appointment, records, consent, bills, payments.
- **Front-Desk Staff:** Registration, check-in, scheduling, queueing.
- **Clinicians (Doctors, Nurses, Therapists):** Notes, orders, care plans, discharge.
- **Pharmacists & Storekeepers:** Prescriptions, dispensing, stock management.
- **Lab & Imaging Personnel:** Orders, specimen/study, verification, result release.
- **Insurance & Payers:** Eligibility, pre-authorization, claims adjudication.
- **Finance & Billing:** Tariffs, invoices, settlements, refunds.
- **Supply Chain & Procurement:** Requisitions, purchase orders, vendor performance.
- **Public Health Officials:** Notifiable disease intake, investigations, dashboards.
- **Hospital Leadership:** Operational KPIs, compliance, audit reviews.

(See Appendix A for detailed persona needs and access boundaries.)

6. Value Proposition

- **For Patients:** Faster, safer care; transparent costs; control over consent.
 - **For Providers:** Less rework; fewer errors; predictable schedules; actionable alerts.
 - **For Payers:** Cleaner claims; lower fraud; faster adjudication.
 - **For Government:** Reliable surveillance; equitable access insights; policy-making data.
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7. High-Level Business Processes

1. **Patient Onboarding & Identity Resolution** → Consent capture → Record merge when duplicates are found.
2. **Appointment** → Check-In → **Encounter Management** → Queueing → Discharge.
3. **Clinical Documentation** → Orders (medications, lab, imaging) → Results → Care plan updates.
4. **Medication Fulfillment** → Dispensing → Refill management → Adherence follow-up.
5. **Diagnostics** → Order validation → Specimen/study → Result verification → Critical notifications.
6. **Billing & Payments** → Estimates → Invoicing → Discounts/waivers → Payment plans.
7. **Insurance** → Eligibility → Pre-authorization → Claims → Adjudication → Appeals.
8. **Inventory & Procurement** → Requisitions → Purchase orders → Receipt → Put-away → Consumption → Reorder.

9. **Public Health Reporting** → Case creation → Investigation workflow → Alerts → Outcomes.

10. **Analytics & Management Reporting** → KPIs → Benchmarking → Action logs.

(Non-technical flows; see Appendix B for swimlane narratives.)

8. Detailed Business Requirements

Notation: Requirements are grouped by domain with IDs like **BR-PA-####**. Each includes: **Objective, Triggers, Actors, Standard Flow, Exceptions, Business Rules, Notifications, Timelines/SLAs, Reports, Acceptance Criteria.**

8.1 Patient Administration (PA)

Objective: Establish accurate, consent-aware patient records and resolve duplicates.

- **BR-PA-001 (Create/Update Patient):**

Trigger: New patient arrival or demographic change.

Actors: Patient/Caregiver, Front-Desk.

Rules:

- Mandatory fields: full name, date of birth, sex at birth, government ID (if available), primary contact, address, next of kin.
- Capture multiple identifiers (national, facility, insurer).
- Consent categories (care, research, communication) selectable individually; default is **care only** if not specified.

Exceptions: No ID → issue temporary registration with reconciliation within **2 business days**.

Notifications: Confirmation SMS/email to patient on new registration/changes.

Acceptance: Record saved with time-stamped audit entry; confirmation provided; consent status visible on patient header.

- **BR-PA-002 (Duplicate Detection & Merge):**

Trigger: Potential duplicate detected during registration/search.

Rules: Merge requires **two-person review** (Front-Desk + Supervisor) with documented rationale; original records remain referenced in a merge log.

Acceptance: Post-merge: single active profile; historical references preserved; all linked encounters visible under the surviving profile.

- **BR-PA-003 (Consent Management):**

Rules: Consent can be granted, limited, or withdrawn by the patient; emergency override requires senior clinician justification and is time-boxed to the encounter.

Acceptance: Access decisions honor consent; override events are flagged in privacy audits.

8.2 Encounter & Scheduling (EM)

Objective: Ensure timely access to care and clear lifecycle management of visits.

- **BR-EM-010 (Scheduling):**

Rules: Slot allocation respects provider availability, specialty, room/equipment needs, and max daily load; enforce overbooking limits per department policy.

Exceptions: Double-book only with Department Head approval; auto-notify impacted patients.

KPIs: Appointment no-show rate tracked by clinic; reschedule lead time.

- **BR-EM-011 (Check-In & Queueing):**

Rules: Priority for urgent cases, elderly, pregnant, and special needs per policy; queue display anonymized.

Acceptance: Average waiting time reported at day/clinic level; patient receives current position estimate.

- **BR-EM-012 (Admission/Transfer/Discharge):**

Rules: Admission requires bed and consultant assignment; transfer requires destination acceptance; discharge requires summary and medication reconciliation.

Timelines: Discharge summary available within **2 hours** post discharge.

Acceptance: Each state change is visible in encounter history and appears in operational dashboards.

8.3 Clinical Documentation (CD)

Objective: Accurate, timely, and auditable clinical records.

- **BR-CD-020 (Structured Notes):**

Rules: Standard templates (SOAP, operative, discharge) per specialty; mandatory sections cannot be bypassed.

Acceptance: Finalized notes are locked; addendums permitted with reason; version history retained.

- **BR-CD-021 (Coding):**
Rules: Diagnoses and procedures coded using nationally adopted classifications; system validates incompatible code pairs; provisional vs confirmed diagnoses distinguished.
Acceptance: Coding completeness $\geq 95\%$ for discharges within **24 hours**.
- **BR-CD-022 (Clinical Alerts):**
Rules: Alert policies for vitals trends, fall-risk, sepsis bundles; alerts must be acknowledged.
Acceptance: Acknowledgment timestamps visible; unacknowledged alerts escalate to charge nurse after **15 minutes**.

8.4 Medication Management (MM)

Objective: Safe prescribing, dispensing, and adherence.

- **BR-MM-030 (E-Prescription):**
Rules: Must specify dose, route, frequency, duration, and indication; check for allergies and interactions; pediatric dosing requires weight-based calculation entry.
Exceptions: Override requires clinician justification and creates a safety flag.
Acceptance: Patient receives medication instructions; pharmacy queue updated.
- **BR-MM-031 (Dispensing & Refill):**
Rules: Dispense against active prescriptions only; substitutions per formulary policy; controlled substances require dual verification.
KPIs: Dispensing turnaround times; stockout incidents.
Acceptance: Label includes patient name, directions, and warnings; inventory decremented; lot/expiry captured.
- **BR-MM-032 (Medication Reconciliation):**
Rules: At admission and discharge, reconcile home meds vs in-hospital meds; discrepancies recorded.
Acceptance: Discharge summary includes reconciled list and patient education.

8.5 Diagnostics — Laboratory & Imaging (DG)

Objective: Complete, timely, and traceable diagnostics.

- **BR-DG-040 (Order Validation):**
Rules: Verify test prerequisites (e.g., fasting), specimen type, priority; prevent duplicates within defined windows.

Acceptance: Orders carry clear collection instructions; conflicting orders flagged.

- **BR-DG-041 (Specimen/Study Workflow):**

Rules: Barcoded chain-of-custody checkpoints (collected, received, in-process, verified); imaging studies include preparation and post-procedure notes.

Timelines: Routine labs **within 6 hours**, stat **within 60 minutes**; imaging report for urgent cases **within 2 hours** of study.

Acceptance: Complete trace history visible; lost/damaged specimen procedure documented.

- **BR-DG-042 (Result Verification & Critical Escalation):**

Rules: Technologist verification followed by pathologist/radiologist sign-off; critical results escalate to responsible clinician and charge nurse until acknowledged.

Acceptance: 100% critical results acknowledged ≤ 30 minutes.

8.6 Insurance & Claims (IC)

Objective: Reduce rejections and accelerate payment.

- **BR-IC-050 (Eligibility & Coverage):**

Rules: Verify coverage and co-pay before high-cost services; capture pre-authorization approvals and reference numbers.

Acceptance: Estimate of patient responsibility provided prior to service.

- **BR-IC-051 (Claim Creation & Submission):**

Rules: Compile clinical, financial, and authorization data; apply payer-specific rules and documentation; support multi-payer splits.

Exceptions: Missing documentation triggers a checklist and blocks submission.

Acceptance: Submission confirmation with tracking; first-pass acceptance $\geq 90\%$ by month 12.

- **BR-IC-052 (Appeals & Denials):**

Rules: Standardized denial reasons; appeal deadlines tracked; attachments managed.

Acceptance: Appeal success rate and cycle time reported per payer.

8.7 Billing & Payments (BP)

Objective: Transparent, accurate patient and payer billing.

- **BR-BP-060 (Tariffs & Estimates):**

Rules: Charges derived from service catalog, packages, and discounts; one price list active per facility at a time; compassionate waivers require review board approval.

Acceptance: Patients receive itemized estimates and final bills; discrepancies explained.

- **BR-BP-061 (Invoicing & Receipts):**

Rules: Invoices reflect services rendered; deposits and advances adjusted; refunds processed with approval trail.

Acceptance: Receipt issued instantly upon payment; patient billing history accessible.

- **BR-BP-062 (Payment Plans):**

Rules: Installments with defined schedule, interest/penalties as per policy; missed payments trigger reminders and alternate arrangement offers.

Acceptance: Plan status viewable by Finance and patient; delinquency escalations logged.

8.8 Inventory & Procurement (IV)

Objective: Ensure drug and consumable availability with minimal waste.

- **BR-IV-070 (Stock Accuracy & Traceability):**

Rules: Record lot/expiry and manufacturer for all receipts; FEFO (first-expiry-first-out) dispensing; quarantine expired or recalled items.

Acceptance: Monthly stock variance $\leq 1\%$ by value.

- **BR-IV-071 (Replenishment & Forecasting):**

Rules: Reorder points derived from consumption and seasonality; emergency procurement path defined.

Acceptance: Stockout days per critical drug $\leq 2\%$ annually.

- **BR-IV-072 (Vendor Management):**

Rules: Approved vendor lists; performance scorecards (on-time delivery, quality, price adherence).

Acceptance: Quarterly vendor performance reviews.

8.9 Public Health Surveillance (PH)

Objective: Timely, accurate national reporting and outbreak response.

- **BR-PH-080 (Notifiable Disease Cases):**

Rules: Mandatory case creation when qualifying lab/imaging/clinical criteria met; local validation; automatic routing to national registry.

Timelines: Report within **60 minutes** of confirmation.

Acceptance: Case ID returned; investigation assignment recorded.

- **BR-PH-081 (Contact Tracing & Alerts):**

Rules: Close-contact identification rules per disease; exposure notifications via approved channels.

Acceptance: Exposure lists maintained; alert delivery status tracked.

- **BR-PH-082 (Immunization Registry):**

Rules: Record vaccine type, lot, site, and adverse events; schedule reminders for follow-up doses.

Acceptance: Coverage dashboards by district.

8.10 Analytics, Management Reporting & Dashboards (AN)

Objective: Operational transparency and strategic insight.

- **BR-AN-090 (Operational KPIs):**

Examples: Waiting time, bed occupancy, theater utilization, claim status, stockouts, critical alert acknowledgments.

Acceptance: Standard dashboards per role; drill-downs to clinic/ward.

- **BR-AN-091 (Ad-Hoc Queries & Cohorts):**

Rules: Authorized users can define cohorts for quality improvement and policy evaluation using de-identified data.

Acceptance: Query results exportable per data governance rules.

9. Roles, Access & Oversight (Business View)

- **Role Families:** Patient, Caregiver, Front-Desk, Clinician, Nurse, Pharmacist, Lab Tech, Radiologist, Billing, Finance, Insurance Assessor, Storekeeper, Procurement Officer, Public Health Officer, Facility Admin, National Admin, Auditor.

- **Principles:** Minimum necessary access, consent-aware access, emergency override with justification and audit, segregation of duties (e.g., billing vs waiver approval).

10. Policy, Compliance & Ethics (Business)

- Adhere to national health data protection and patient rights legislation.
 - Explicit patient consent for non-care uses (e.g., research communications).
 - Sensitive events (mental health, reproductive health, communicable diseases) require enhanced oversight.
 - Retention schedules: clinical and financial records per statutory timelines; legal hold procedures defined.
 - Audit trails for all sensitive actions; periodic audit reviews with findings and remediation.
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11. Service Levels & KPIs (Business)

- **Availability of front-line services:** Clinic hours; planned downtime notifications ≥ 72 hours in advance.
 - **Turnaround Times:** Admissions ≤ 15 min after bed availability; routine labs ≤ 6 h, stat ≤ 60 min; discharge summary ≤ 2 h.
 - **Financial:** First-pass claim acceptance $\geq 90\%$; average days in AR ≤ 45 .
 - **Inventory:** Stockout days of critical drugs $\leq 2\%$; expired items quarantined same day.
 - **Public Health:** Notifiable reporting on time $\geq 95\%$.
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12. User Experience & Accessibility (Business)

- Clear, plain-language interfaces; local language support; large-font options.
- Queue status visibility without exposing identities.

- Patient bills and instructions written at an accessible reading level; pictograms for dosing.
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13. Reporting Catalogue (Examples)

- **Clinical:** Sepsis bundle compliance, readmission within 30 days, surgical site infections.
 - **Operational:** Appointment no-show, average wait, bed occupancy, OT utilization.
 - **Financial:** Revenue by service line, payer mix, write-offs, refund turnaround.
 - **Inventory:** ABC analysis, stock aging, vendor performance.
 - **Public Health:** Case counts by district, R(t) proxy, immunization coverage.
(Complete catalogue in Appendix C with owners, frequency, and distribution lists.)
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14. Implementation Phasing (Business View)

- **Phase 1 (Months 0–6):** Patient administration, scheduling, outpatient clinical documentation, basic billing, pharmacy, routine labs, foundational reporting.
 - **Phase 2 (Months 6–12):** Inpatient flows, imaging, advanced billing, insurance claims, inventory & procurement.
 - **Phase 3 (Months 12–18):** Public health registries, patient portal at scale, advanced analytics, appeals & denials, capacity planning.
Each phase includes change management, training, and UAT cycles with business sign-off.
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15. Training, Communication & Change Management

- **Training:** Role-based curricula; scenario practices (e.g., emergency admission); super-user program per facility.

- **Communication:** Go-live readiness scorecards; patient-facing FAQs; helpline staffing.
 - **Adoption Metrics:** Login rates, documentation completeness, alert acknowledgments, billing accuracy.
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16. Risks & Mitigations

- **Data Quality (duplicates, miscoding):** Data stewardship roles; merge governance; coding audits.
 - **Operational Disruption at Go-Live:** Parallel runs; on-site floor support; contingency paper flows.
 - **Stakeholder Resistance:** Early involvement; clinical champions; feedback loops.
 - **Policy Variance Across Facilities:** Standard operating procedures with controlled local variations.
 - **Timely Public Health Reporting:** Clear responsibility matrix; escalation rules.
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17. Assumptions & Dependencies

- National identity, provider, drug, and disease registries are maintained by respective authorities.
 - Facilities designate data stewards and super-users.
 - Payers provide current coverage and authorization rules and SLA commitments.
 - Legal and privacy frameworks enable the specified uses with consent.
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18. Business Acceptance Criteria (UAT-Level)

- **BAC-01 Patient Lifecycle:** Create, update, merge, and consent flows validated with 25 real-world scenarios, including emergency override and reconciliation.
 - **BAC-02 Encounter Flows:** 30 end-to-end journeys from appointment to discharge across specialties with time checks.
 - **BAC-03 Diagnostics:** 20 lab and 10 imaging scenarios including critical value escalations and lost specimen handling.
 - **BAC-04 Pharmacy:** 25 prescriptions including pediatric dosing, substitution, controlled substances.
 - **BAC-05 Billing & Claims:** 30 scenarios covering estimates, waivers, multi-payer claims, appeals.
 - **BAC-06 Inventory:** 20 stock scenarios including FEFO, quarantines, recalls, emergency procurement.
 - **BAC-07 Public Health:** 15 notifiable case scenarios including contact tracing and alerting.
 - **BAC-08 Reporting:** Validate 30 priority reports and 10 dashboards for correctness and timeliness.
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19. Traceability (Goals → Requirements → Measures)

Goal	Representative Requirements	Measures
G1	BR-PA-001/2/3	% successful patient retrieval; merge backlog days
G2	BR-EM-010/11/12	Avg wait time; discharge summary timeliness
G3	BR-DG-042; BR-CD-022	Critical acknowledgment time; alert escalation counts
G4	BR-IC-051/52; BR-BP-060/61	First-pass acceptance; days in AR
G5	BR-PH-080/81	On-time reporting rate; investigation start times

G6	All "audit-sensitive" BRs	Completed audit review cycles; exceptions closed
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